QSAR + REINVENT Pipeline Overview
Simple, Process-Focused Project Presentation

Based on the comprehensive report and repository structure documents

QSAR Project Team

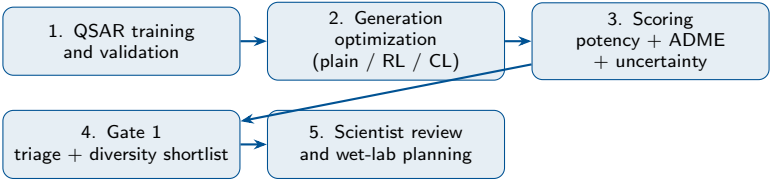
Computational Drug Discovery

March 16, 2026

What this presentation covers

1. QSAR to translation walkthrough
2. QSAR Modeling (Detailed)
3. REINVENT4 Generation (Detailed)
4. Gate 1 (Detailed)
5. Gate 2 / Gate 3 (Legacy, Inactive)
6. Why this project
7. Key terms (plain language)
8. REINVENT 4 in this project
9. Process schema
10. How scoring is carried out (math)
11. Repository map and outputs
12. Practical usage
13. Excel field guide
14. Tools used in this repository

Pipeline order used in this project



Why this ordering

Learn from known data first (QSAR), optimize generation next, then use Gate 1 triage and diversity control before selecting compounds for lab planning.

Data acquisition → deployment lifecycle

1) Data acquisition

Experimental series data (C/E + historical labels), SMILES cleaning, canonicalization, and deduplication.

2) Modeling

Descriptor generation + model selection for potency prediction. Best potency model: **RDKit + KNN**.

3) Evaluation

Held-out validation on Series C and prospective ranking on Series E.

4) Testing (in-silico pipeline testing)

REINVENT generation campaigns, automated scoring, and Gate 1 shortlist construction for downstream review.

5) Deployment / operational use

Final ranked workbook is produced for chemistry/biology teams to choose wet-lab candidates and plan experiments.

Key current results

Series C: MAE **0.118**, $R^2 = \mathbf{0.887}$
Gate 1 shortlist: **200** candidates

Step 1: QSAR model training and key result

How QSAR is built

1. Clean and standardize molecular structures.
2. Compute descriptor families (RDKit/ECFP/MACCS/Mordred).
3. Train multiple regressors and compare on held-out data.
4. Keep best model for primary IC_{50} prediction.

Key performance (current artifact)

Best model:
RDKit + KNN
$R^2 \approx 0.887$

Additional positive-R^2 models used for uncertainty:

- ECFP_r3 + KNN
- RDKit + Decision Tree
- RDKit + LGBM
- RDKit + Extra Trees

Where QSAR test outputs are saved (Series C/E)

Primary C/E prediction files

`outputs/qsar/core_predictions/SeriesC_test_with_predictions.csv`

`outputs/qsar/core_predictions/SeriesE_with_predictions.csv`

Manifest files

`outputs/qsar/manifests/qsar_outputs_manifest.csv`

`outputs/qsar/manifests/additional_qsar_outputs_manifest.csv`

What each contains

- Canonical structures and identifiers
- Predicted pIC_{50} and predicted IC_{50} (μM)
- Source/series metadata and downstream analysis fields

Series C and E testing: key QSAR outputs

Series C (test set)

- Total rows: **6**
- Rows with measured pIC_{50}: **5**
- MAE (pIC_{50}): **0.118**
- R^2: **0.887**

Series E (prediction set)

- Total rows predicted: **6**
- Predicted pIC_{50} range: **4.467 to 5.132**
- Predicted IC_{50} range: **7.38 to 34.11 μM**

Interpretation for presentation

Series C confirms model quality on known test compounds; Series E provides prospective potency ranking for compounds without direct fitted labels.

Step 2: Generation optimization (plain, RL, CL)

How optimization works

- **Plain**: broad exploration from prior model.
- **RL**: sample → score → update agent.
- **CL**: staged objectives from easier to stricter targets.

Why CL is important

More stable learning, fewer invalid molecules, better quality at convergence.

Used in this project

Families/modes were run across campaigns, then all sampled outputs were passed through one consistent scoring and gating workflow.

Typical path:

`reinvent_integration/results/`

`<family_mode>/samples.csv`

Step 3: Scoring and ranking logic

Scoring stack

Validate SMILES → descriptors → potency prediction → ADME/risk enrichment → uncertainty estimation → gate-ready output fields.

Gate 1 triage equation

$$\text{triage} = 0.35P + 0.15S + 0.15L + 0.10M + 0.10H + 0.15T$$

P: potency, S: solubility, L: cLogP window, M/H: clearance terms, T: half-life.

Scored output location

`outputs/generated/<family>/<mode>/`
CSV + Excel with structures and fields used by gates.

How uncertainty is controlled in ranking

Uncertainty calculation

Up to 5 positive-R^2 models predict each molecule.

$$\mu = \frac{1}{k}\sum_{i=1}^{k}\hat{y}_i, \quad \sigma = \sqrt{\frac{1}{k-1}\sum_{i=1}^{k}(\hat{y}_i - \mu)^2}$$

Lower σ means higher confidence.

Control mechanism

- Primary potency remains best model output.
- Uncertainty columns are retained for analysis and confidence review.
- Confidence columns are carried into final ranking.

Practical effect

Reduces false positives that look strong in one model but unstable across the ensemble.

Step 4: Active shortlist output (Gate 1)

What Gate 1 optimizes

- Potency + ADME multi-objective triage
- Scaffold-diverse shortlist selection

Current run note

`pred_pIC50_uncertainty` = 0 in the current scored outputs, indicating a single effective model is being used for potency prediction. Ensemble uncertainty is available when multiple positive-R^2 models are active during scoring.

- Consistent ranking fields for review
- Decision-ready workbook export

Current outputs

`outputs/hybrid_stage/gate1/gate1_shortlist.xlsx`

Current count

Gate 1 shortlist: **200**

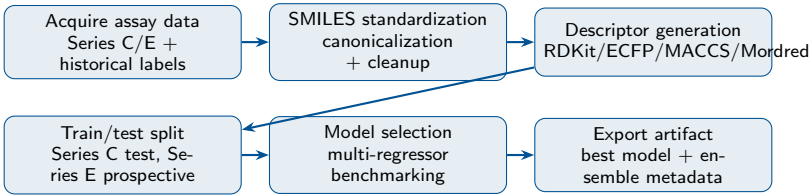
What was done and what you can present

1. Built QSAR with best-performing potency model and ensemble uncertainty.
2. Ran REINVENT generation campaigns across families and optimization modes.
3. Scored all generated molecules with one consistent scoring pipeline.
4. Applied Gate 1 triage ranking and scaffold-diverse shortlist selection.
5. Produced ranked Excel outputs for medicinal-chemistry review and lab planning.

Final narrative

QSAR learns, **REINVENT explores/optimizes**, and **Gate 1 delivers an actionable shortlist**.

QSAR modeling flowchart: data acquisition to model artifact



QSAR modeling details: what is trained and selected

Candidate models tested

- KNN, Decision Tree, Extra Trees
- Gradient boosting family (e.g., LGBM)
- Descriptor families compared side by side
- Selection by held-out predictive quality

Selected setup (current)

- Primary potency model: **RDKit + KNN**
- Key metric: $R^2 \approx 0.887$
- Backup positive-R^2 models kept for uncertainty
- Artifact path:

  `reinvent_integration/artifacts/qsar_best_model.joblib`

QSAR evaluation and test outputs (Series C and E)

Series C (test evaluation)

- Total rows: **6**
- Measured pIC_{50} rows: **5**
- MAE (pIC_{50}): **0.118**
- R^2: **0.887**

Series E (prospective predictions)

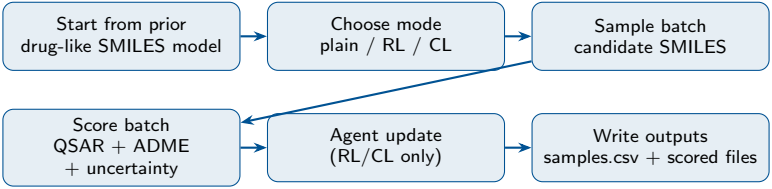
- Predicted rows: **6**
- Predicted pIC_{50}: **4.467 to 5.132**
- Predicted IC_{50}: **7.38 to 34.11 μM**

Output files

`outputs/qsar/core_predictions/SeriesC_test_with_predictions.csv`

`outputs/qsar/core_predictions/SeriesE_with_predictions.csv`

REINVENT4 generation flowchart



REINVENT4 modes and optimization details

Mode	How optimization behaves
Plain	No policy update; broad exploration from prior distribution.
RL	Iterative policy optimization from reward signal after scoring each batch.
CL	Multi-stage objective: easier constraints first, stricter objectives later.
Mol2Mol/LibINVENT/LinkINVENT	Family-specific generation constraints layered on top of training mode.

Why curriculum learning was used

It improves stability and validity while pushing toward higher-quality, multi-objective molecules under realistic constraints.

REINVENT4 project execution and outputs

What was run

- Multiple generation families and modes
- Automated score pass after each sampling output
- Unified downstream gate ranking across all runs

Where outputs are saved

Raw samples:

`reinvent_integration/results/<family_mode>/samples.csv`

Scored results:

`outputs/generated/<family>/<mode>/`

Generation activity scope in this project (not just 3 modes)

Model families orchestrated in campaigns

reinvent, reinvent_pubchem, pepinvent, mol2mol_similarity, mol2mol_high_similarity, mol2mol_medium_similarity, mol2mol_scaffold, mol2mol_scaffold_generic, mol2mol_mmp, libinvent, libinvent_transformer_pubchem, linkinvent, linkinvent_transformer_pubchem, and the nitro_bioisostere_campaign.

Operational implication

Each family was executed across plain/RL/CL flows with post-scoring, so the generation program is a **matrix of campaigns**, not a single run. Current generation coverage is fully populated across the targeted branches.

Generation campaign matrix: families × optimization modes

Family group	Plain	RL	CL
REINVENT / REINVENT_PubChem	✓	✓	✓
PepINVENT	✓	✓	✓
Mol2Mol (sim/high/medium/scaffold/scaffold_generic/mmp)	✓	✓	✓
LibINVENT (+ transformer prior)	✓	✓	✓
LinkINVENT (+ transformer prior)	✓	✓	✓
Nitro bioisostere campaign	✓	✓	✓

Why this matrix matters

It separates exploration (plain), exploitation (RL), and robust staged optimization (CL), while covering multiple molecular design tasks (de novo, analoging, linking, decorating).

REINVENT / REINVENT_PubChem (✓✓✓)

What this family contributes

- Core de novo generation backbone
- REINVENT_PubChem adds broader prior chemistry exposure
- Strong for novelty-first exploration

Modes in this project

Plain: baseline de novo sampling
RL: reward-guided potency/ADME optimization
CL: staged objective tightening for stable convergence

Concrete run artifacts

Config examples:

`sample_reinvent_plain.toml`

`rl_reinvent.toml` / `cl_reinvent.toml`

`sample_reinvent_rl.toml` / `sample_reinvent_cl.toml`

Raw outputs:

`reinvent_integration/results/reinvent_<mode>/samples.csv`

Scored outputs:

`outputs/generated/reinvent/<mode>/`

PepINVENT (✓✓✓)

What this family contributes

- Peptide-like sequence/structure generation focus
- Complements small-molecule-focused families
- Adds modality diversity to candidate pool

Modes in this project

Plain: broad peptide-like proposal space
RL: prioritize higher reward peptide candidates
CL: staged constraints to improve quality stability

Concrete run artifacts

Config examples:

`sample_pepinvent_plain.toml`

`rl_pepinvent.toml` / `cl_pepinvent.toml`

`sample_pepinvent_rl.toml` / `sample_pepinvent_cl.toml`

Raw outputs:

`reinvent_integration/results/pepinvent_<mode>/samples.csv`

Scored outputs:

`outputs/generated/pepinvent/<mode>/`

Mol2Mol (sim/high/medium/scaffold/mmp) (✓✓✓)

What this family contributes

- Controlled analog generation around references
- Multiple transformation regimes: similarity, scaffold, MMP
- Balances novelty with medicinal chemistry continuity

Modes in this project

Plain: analog exploration without policy updates
RL: optimize transformations toward reward
CL: staged analog objective tightening per regime

Concrete run artifacts

Examples:

`sample_mol2mol_similarity_plain.toml`

`rl_mol2mol_high_similarity.toml`

`cl_mol2mol_scaffold.toml`

Raw outputs:

`reinvent_integration/results/mol2mol_<variant>_<mode>/s`

Scored outputs:

`outputs/generated/mol2mol/<variant>/<mode>/`

LibINVENT (+ transformer prior) (✓✓✓)

What this family contributes

- Scaffold decoration/library-style expansion
- Useful for R-group exploration at fixed cores
- Transformer prior variant broadens syntactic diversity

Modes in this project

Plain: enumerate baseline decoration ideas
RL: improve decoration reward profile
CL: staged decoration quality constraints

Concrete run artifacts

Config examples:

`sample_libinvent_plain.toml`

`rl_libinvent.toml` / `cl_libinvent.toml`

`sample_libinvent_transformer_pubchem_cl.toml`

Raw outputs:

`reinvent_integration/results/libinvent_<mode>/samples.csv`

`reinvent_integration/results/libinvent_transformer_pubchem`

LinkINVENT (+ transformer prior) (✓✓✓)

What this family contributes

- Linker design between anchors/fragments
- Critical for fragment-joining and geometry-aware assembly
- Transformer prior variant increases linker novelty breadth

Modes in this project

Plain: linker idea exploration
RL: optimize linker candidates by reward
CL: staged linker constraint tightening

Concrete run artifacts

Config examples:

`sample_linkinvent_plain.toml`

`rl_linkinvent.toml` / `cl_linkinvent.toml`

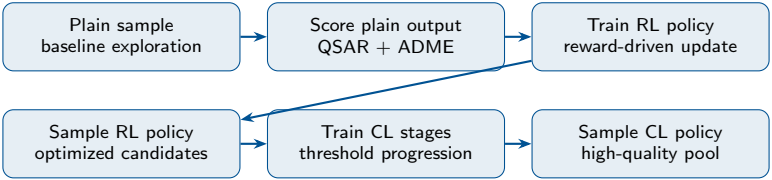
`sample_linkinvent_transformer_pubchem_cl.toml`

Raw outputs:

`reinvent_integration/results/linkinvent_<mode>/samples.cs`

`reinvent_integration/results/linkinvent_transformer_pubch`

Generation execution lifecycle (per family)



How run orchestration is implemented

Controller scripts

`reinvent_integration/run_all_generation.py`
`reinvent_integration/run_all_generation_fast.py`
`reinvent_integration/run_all_generation_strict.py`
`reinvent_integration/run_nitro_bioisostere_campaign.py`

Automated per-family sequence

For each family: plain sample → score, RL train → RL sample → score, CL train → CL sample → score, then merge into hybrid stage-gate pipeline.

Diverse generation activities: family-specific intent

Exploration and analoging

- REINVENT / REINVENT_PubChem: broad de novo novelty
- Mol2Mol similarity/high/medium: controllable analog search
- Mol2Mol scaffold/scaffold_generic: scaffold-preserving design
- Mol2Mol MMP: matched-pair style transformations

Assembly and decoration tasks

- PepINVENT: peptide-like sequence/structure design
- LinkINVENT: linker optimization between anchors
- LibINVENT: scaffold decoration libraries
- Transformer-PubChem variants: broader prior transfer

Optimization depth: what RL and CL improve over plain

Aspect	Practical behavior in pipeline
Potency targeting	RL/CL shift sampled distribution toward higher QSAR-ranked molecules.
ADME alignment	Objective shaping increases fraction within preferred ADME windows.
Feasibility control	CL staging reduces abrupt jumps that produce unstable candidates.
Gate readiness	RL/CL outputs generally show better downstream Gate 1 shortlist quality.

Generation run variants: default, fast, strict, campaign-specific

Default/full campaign

Broad model matrix, full scoring stack, standard gate thresholds for ranked export.

Strict variant

Tighter constraints / stricter progression, useful for high-confidence candidate narrowing.

Fast variant

Reduced runtime profile for rapid iteration and sanity checks.

Nitro bioisostere campaign

Focused chemistry campaign with dedicated configs and logs.

Generation logs: what information is captured

Log locations and examples

`reinvent_integration/logs/run_all_generation.log`
`reinvent_integration/logs/run_all_generation_fast.log`
`reinvent_integration/logs/run_all_generation_strict.log`
Family logs: `<family>_<mode>_<train/sample>_*.log`

Operational signals extracted

- step completion/failure status for each run
- training progress and sampling completion
- mode-specific checkpoints and campaign diagnostics

Generation output hierarchy and handoff to scoring

Raw generation outputs

`reinvent_integration/results/<family>_<mode>/sa`
plus campaign-level artifacts/checkpoints in results tree.

Scored generation outputs

`outputs/generated/<family>/<mode>/`
standardized scored files used by Gate 1 input pooling.

End-to-end linkage

Generation activity → scoring normalization → Gate 1 ranking → scientist review and wet-lab planning.

REINVENT4 generation families: what each generates

Family	Generation behavior and practical use
REINVENT (de novo)	Generates full molecules from learned prior; best for broad exploration.
Mol2Mol	Starts from reference scaffold/seed and proposes close-but-improved analogs.
LinkINVENT	Generates linkers to connect fragments while preserving anchor context.
LibINVENT	Enumerates decoration vectors around a scaffold under reaction-like constraints.
Transformer-PubChem variants	Broader pretraining bias for novelty and chemical grammar robustness.

Project strategy

Use multiple families in parallel: broad exploration (de novo) + local optimization (scaffold/fragment-driven families), then merge all outputs into one scoring/gating path.

Plain generation details (exploration mode)

What happens in plain mode

- Agent weights are fixed (no optimization loop)
- Batched sampling from prior policy
- Strongly useful for baseline diversity
- Fast throughput for first-pass idea space

How outputs are used

- Post-scored by QSAR + ADME pipeline
- Feeds Gate 1 diversity-aware shortlist
- Acts as discovery baseline vs RL/CL gains

Reinforcement learning generation: optimization loop

Sample molecules from current policy → Compute reward multi-objective score → Update policy increase expected reward → Resample and repeat until step budget

Optimization objective (conceptual)

$$J(\theta) = \mathbb{E}_{x \sim \pi_\theta}[R(x)] \quad \Rightarrow \quad \theta \leftarrow \theta + \eta \nabla_\theta J(\theta)$$

Where $R(x)$ combines potency, ADME, and feasibility/risk constraints.

Curriculum learning generation: staged optimization

Why curriculum is stronger than single-stage RL

1. Stage 1: satisfy validity/basic chemistry and minimal potency.
2. Stage 2: tighten potency + key ADME windows.
3. Stage 3: enforce full objective stack before final sampling.

This reduces optimization collapse and improves final feasible hit-rate.

Progression rule (conceptual)

Advance stage when:

$$\bar{R}_{\text{window}} \geq \tau_s$$

where τ_s is stage threshold.

Reward construction used for REINVENT optimization

Composite reward structure

$$R = w_p f_{\text{potency}} + w_s f_{\text{solubility}} + w_l f_{\log P} + w_m f_{\text{microsomal}} + w_h f_{t_{1/2}} - w_u f_{\text{uncertainty}} - w_t f_{\text{tox risk}}$$

Positive terms drive efficacy/developability; penalty terms suppress unstable or risky candidates.

Important practical point

Generation optimization and gate triage are aligned but not identical: reward guides learning dynamics, gates enforce strict deployment filters.

How to read REINVENT optimization quality in this project

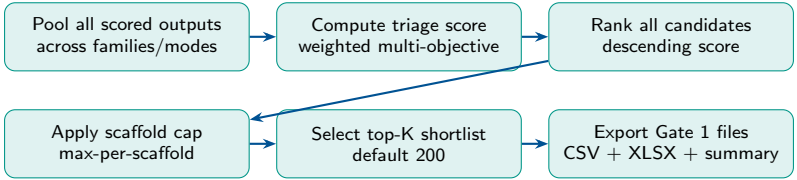
Healthy optimization signals

- Mean score trend increases over steps
- Valid SMILES fraction stays high
- Diversity does not collapse too early
- Top candidates remain competitive in Gate 1 shortlist ranking

Warning patterns to monitor

- Score spike with low diversity (mode collapse)
- Rising uncertainty in high-score molecules
- ADME out-of-window drift at later steps
- Weak Gate 1 shortlist enrichment ratio

Gate 1 flowchart: triage and diversity shortlist



Gate 1 details: scoring and output artifacts

Triage equation

$$\text{triage} = 0.35P + 0.15S + 0.15L + 0.10M + 0.10H + 0.15T$$

Potency is weighted highest; ADME balance and stability terms control quality.

Gate 1 outputs

`outputs/hybrid_stage/gate1/gate1_master_ranked.csv`

`outputs/hybrid_stage/gate1/gate1_shortlist.csv`

`outputs/hybrid_stage/gate1/gate1_shortlist.xlsx`

`outputs/hybrid_stage/gate1/gate1_summary.json`

Gate 1 shortlist: candidate pool composition (current)

Source family breakdown (200 shortlisted)

Family	Count
reinvent_pubchem	137
nitro_bioisostere_campaign	53
reinvent	9
linkinvent_transformer_pubchem	1

Source mode breakdown

Mode	Count
plain	140
mol2mol_similarity	28
mol2mol_scaffold_generic	22
curriculum_learning	6
mol2mol_scaffold	3
reinforcement_learning	1

Observation

Plain-mode candidates dominate (70%), with mol2mol analog variants contributing a significant diversity-preserving portion (26.5%). Master ranked pool: **37,451** candidates from **14,287** unique scaffolds.

Gate 2 and Gate 3 status in current pipeline

Inactive stages

Gate 2 and Gate 3 are removed from the active implementation in `run_hybrid_stage_gates.py`. Gate 1 is the only production stage.

How to interpret old references

- Existing Gate 2/3 descriptions in legacy notes are historical context.
- Current ranked outputs are written only under `outputs/hybrid_stage/gate1/`.
- Legacy folders remain for provenance in `outputs/hybrid_stage/legacy/`.

Problem we are solving

Goal

Start from many AI-generated molecules and end with a small, high-quality shortlist for laboratory testing.

- We do **not** have a reliable protein target structure for physics-heavy simulation.
- So we use a **target-free, data-driven pipeline**.
- We combine potency prediction, ADME quality, uncertainty, safety risk, and diversity before selecting final candidates.

Business rationale

Reduce expensive false positives early, while keeping chemical diversity high.

Core idea in one line

Generate many molecules $\rightarrow$ score quickly $\rightarrow$ rank consistently $\rightarrow$ prioritize for lab review

Key terms everyone should know

Term	Simple meaning
SMILES	Text format for molecules (a way to write chemistry as strings).
QSAR	Model that predicts activity/properties from structure.
IC_{50}	Potency: lower value means stronger molecule.
pIC_{50}	Log version of IC_{50}: higher value means stronger molecule.
ADME	Absorption, distribution, metabolism, excretion properties.
Murcko scaffold	Core ring skeleton used to track diversity.
Uncertainty	How much multiple good models disagree on prediction.

Generation terms used in this project

Term	Simple meaning
REINVENT (plain)	Generate molecules directly from pretrained chemistry language model.
RL mode	Reinforcement learning: reward molecules with better predicted profile.
CL mode	Curriculum learning: optimize in staged difficulty.
Mol2Mol	Create analogs from seed molecules.
LibINVENT	Decorate a scaffold with substituents.
LinkINVENT	Connect fragments with designed linker.

How REINVENT 4 generation modes work

Mode	How it generates molecules
Plain (prior sampling)	Samples directly from pretrained chemical language model for broad exploration.
RL (reinforcement learning)	Generates a batch, scores it, then updates the agent to prefer higher-scoring molecules.
CL (curriculum learning)	Runs optimization in stages: easier objective first, then stricter objective later.
Mol2Mol	Starts from known molecules and proposes close analogs.
LibINVENT	Keeps a core scaffold and decorates attachment points with substituents.
LinkINVENT	Connects two fragments with generated linkers.

How curriculum learning optimization is done

Optimization sequence

1. Start from prior model (chemically valid baseline).
2. Stage 1 objective: easier constraints (validity + broad quality).
3. Stage 2 objective: stricter score (potency/ADME/uncertainty).
4. At each step: sample $\rightarrow$ score $\rightarrow$ update agent.
5. Stop at max steps or convergence, then sample final set.

Why curriculum helps

- More stable optimization
- Fewer invalid molecules
- Better quality-diversity tradeoff
- Less mode collapse

How REINVENT 4 was used in this project

What was done

Multiple generation families were run (including Mol2Mol, PepINVENT, LibINVENT, LinkINVENT and transformer variants) in one or more training modes (plain, RL, CL), then each sampled output was scored by the same QSAR + ADME + uncertainty pipeline.

Execution flow

1. Run REINVENT campaign config
2. Write `samples.csv`
3. Score via `score_generated_smiles_adme.py`
4. Save scored CSV/XLSX

Where files are

- Raw generations: `reinvent_integration/results/`
- Scored outputs: `outputs/generated/<family>/<mode>/`
- Gate outputs: `outputs/hybrid_stage/`

REINVENT outputs that feed decision-making

Per-run output

Each campaign produces generated SMILES plus scored tables containing potency, ADME, uncertainty, and source metadata.

Cross-run integration

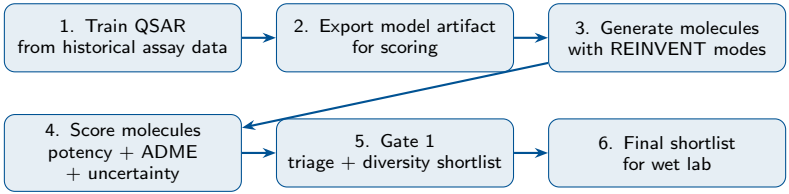
All scored runs are pooled and moved through:

- **Gate 1**: triage + scaffold-diverse top shortlist
- **Active stage**: Gate 1 only in the current workflow
- **Gate 2/3**: legacy reference only (inactive)

Final deliverable

A ranked candidate workbook with structures and rationale-ready fields, used by chemistry and biology teams for lab selection.

End-to-end process schema



Active gate view (simple)

Gate 1

Fast triage

- Potency + ADME
- Composite triage score
- Scaffold-diverse top 200

Current status

Gate 2 and Gate 3 are retained only as historical references in older materials and are not active in the current production workflow.

How scoring is carried out (step-by-step)

1. **Validate SMILES** (remove invalid molecules).
2. **Compute descriptors** from structure.
3. **Predict potency** with best QSAR model.
4. **Estimate uncertainty** from up to 5 positive-R^2 models.
5. **Add ADME / risk fields** (solubility, lipophilicity, clearance).
6. **Apply gate logic** for ranking and filtering.
7. **Export CSV + Excel** with structures for human review.

Outcome

Every candidate has machine-scored evidence + a readable review format, so computational and chemistry teams can make the same decision from the same table.

Scoring mathematics: Gate 1 triage score

Composite triage function

$$\text{triage} = 0.35\,P + 0.15\,S + 0.15\,L + 0.10\,M + 0.10\,H + 0.15\,T$$

with P=potency, S=solubility, L=cLogP window, M=microsomal clearance, H=hepatocyte clearance, T=half-life.

Normalization

Higher-is-better:

$$x_{norm} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Lower-is-better features are inverted after scaling.

Why these weights? (rationale)

- **0.35 Potency**: strongest requirement; non-potent compounds are rarely useful.
- **0.30 Solubility + cLogP**: captures absorption/permeability balance.
- **0.20 Clearance terms**: avoids compounds that disappear too quickly in vivo.
- **0.15 Half-life**: rewards exposure duration without dominating the score.

Important interpretation

This is a **triage utility score**, not a perfect biological truth. It is designed to rank candidates consistently for decision-making.

How uncertainty is computed

Model strategy

Keep the **best $\mathbf{IC}_{50}$** model as primary prediction, then evaluate up to **five positive-R^2** models for the same molecule.

Uncertainty proxy

Let predictions be $\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_k$, $k \leq 5$.

$$\mu = \frac{1}{k}\sum_{i=1}^{k}\hat{y}_i, \quad \sigma = \sqrt{\frac{1}{k-1}\sum_{i=1}^{k}(\hat{y}_i - \mu)^2}$$

Larger σ means lower confidence.

Why it helps

Prevents over-prioritizing molecules that look strong in only one model.

Who carries out scoring? (roles in process)

Role	What they do
REINVENT engine	Generates candidate SMILES across modes (plain/RL/CL).
QSAR scoring script	Computes potency + ADME + uncertainty fields.
Gate script	Applies triage math and diversity selection for Gate 1 shortlist generation.
Medicinal chemist	Reviews computed outputs for structural quality and synthesis practicality.
DMPK/safety scientist	Interprets computed tox/DDI/absorption risk outputs for go/no-go decisions.
Project lead	Approves final shortlist for experiment.

Repository structure (current)

Code folders

- `qsar_core/` core modules
- `reinvent_integration/` generation + scoring + gates
- `scripts/` runnable scripts
- `config/` environment files

Data & outputs

- `data/raw/` input dataset
- `outputs/generated/` scored generations
- `outputs/hybrid_stage/` gate outputs
- `docs/` plain-language documentation

Main output files used in decisions

1) Gate 1 shortlist (diversity + triage)

`outputs/hybrid_stage/gate1/ gate1_shortlist.xlsx`

2) Gate 1 ranked table (full pool)

`outputs/hybrid_stage/gate1/ gate1_master_ranked.csv`

3) Legacy historical outputs (inactive stages)

`outputs/hybrid_stage/legacy/`

How to run the pipeline (practical commands)

From repository root

- Export / refresh model artifact:

  ```
  python reinvent_integration/export_qsar_model.py
  ```

- Run generation pipeline:

  ```
  python scripts/run_pipeline.py
  ```

- Build stage-gated rankings:

  ```
  python reinvent_integration/run_hybrid_stage_gates.py
  ```

Preset options

```
python scripts/run_pipeline.py --generation-preset fast|balanced|diverse
```

Decision logic summary

1. Generate broad chemical space.
2. Score with potency + ADME + uncertainty.
3. Keep diversity while ranking quality.
4. Remove uncertain / high-liability / hard-to-make molecules.
5. Prioritize by Gate 1 triage score, diversity, and experiment value.

Why this works in no-target settings

It is fast, auditable, and aligned with real project decision gates, even when structure-based physics methods are unavailable.

How to read values in the Excel workbooks

Binary / flag fields

- **1** = passes / yes / valid / favorable
- **0** = fails / no / invalid / unfavorable

Active: `passes_adme_individual`, `passes_adme_ensemble`, `passes_optimal`, `passes_metabolic_stability`.
Legacy (inactive): `passes_gate2_target_free`, `passes_gate3_translational`.

Rank fields

- Lower rank number = higher priority
- `rank` = Gate 1 order
- `gate2_rank`, `gate3_rank` = legacy inactive stages

Continuous score fields

- Usually **higher = better** for composite scores
- Usually **lower = safer** for liabilities and uncertainty
- Always interpret with field meaning, not number alone

Key rule of thumb

Higher potency score, higher confidence, higher absorption support are good; **lower IC_{50}, lower uncertainty, lower liability counts, lower toxicity/DDI risk** are good.

Gate 1 workbook fields: structure, potency, ADME, source

Field	Meaning and value interpretation
`Structure` / `canonical_smiles`	Molecule drawing / canonical text structure used for scoring and review.
`SMILES_state`	Validity flag; 1 means structurally accepted, 0 means invalid/problematic.
`pred_pIC50`	Predicted potency on log scale; higher means stronger predicted activity.
`pred_ic50_uM`	Predicted potency in μM; lower means stronger predicted activity.
`pred_cLogP`	Predicted lipophilicity; mid-range is generally preferred over extreme low/high values.
`pred_logS_ESOL`	Predicted solubility log scale; less negative / higher means more soluble.
`pred_MetStab_Clearance_Microsome_AZ`	Microsomal clearance proxy; lower is generally better because slower clearance means better stability.
`pred_MetStab_Clearance_Hepatocyte_AZ`	Hepatocyte clearance proxy; lower is generally better.
`pred_MetStab_Half_Life_Obach`	Metabolic half-life proxy; higher means more stable / longer-lasting.
`passes_optimal`, `passes_adme_individual`, `passes_adme_ensemble`	Binary gate flags; 1 means the molecule is within the preferred rule window.
`source_family`, `source_mode`	Which generator produced the compound and under which campaign mode.

Gate 1 workbook fields: ranking and similarity

Field	Meaning and value interpretation
`rank`	Gate 1 shortlist order; smaller rank means better overall triage priority.
`triage_score`	Main Gate 1 composite score; higher means better combined potency + ADME profile.
`pred_pIC50_uncertainty`	Potency model disagreement; lower means higher confidence.
`Input_SMILES`	Original generator-provided structure string before normalization.
`Tanimoto`	2D structural similarity to the reference/seed context; higher means more similar, lower means more novel.
`murcko_scaffold` / `scaffold_bucket`	Core chemotype grouping used to keep shortlist diversity.
`adme_ensemble_score`	Ensemble ADME quality score combining solubility, lipophilicity, and metabolic stability; higher is better.
`NLL`	Negative log-likelihood from generator; lower typically means the model finds the structure more probable.

Observed Gate 1 behavior in current workbook

`triage_score`: 0.77–0.92; `Tanimoto`: 0.28–0.86; `pred_pIC50`: 4.94–6.11; `pred_ic50_uM`: 0.78–11.55 μM.

Legacy Gate 2 workbook fields (inactive reference)

Field	Meaning and value interpretation
gate2_target_free_score	Gate 2 composite score; higher means stronger confidence + chemistry support.
passes_gate2_target_free	Binary Gate 2 pass flag; 1 means the molecule survives this refinement stage.
pred_pIC50_uncertainty	Ensemble disagreement around potency; lower means more reliable prediction.
pred_pIC50_lower95	Conservative lower confidence bound on potency; higher is better because even the lower bound remains active.
pred_ic50_uM_upper95	Conservative upper confidence bound in μM; lower is better.
uncertainty_confidence_score	Confidence derived from uncertainty; higher means more trust in the potency estimate.
sa_score	Synthetic accessibility score; lower usually means easier to make.
sa_accessibility	Rescaled synthetic accessibility; higher means easier / more accessible synthesis.
liability_alert_count	Count of chemistry alert motifs; lower is better.
liability_free	Binary alert-free flag; 1 means no detected alert motifs.

Legacy Gate 2/3 workbook fields (inactive reference)

Field	Meaning and value interpretation
nearest_active_tanimoto, shape_tanimoto_max	2D/3D support from known actives; higher means stronger ligand-based support.
ligand_support_score	Combined ligand evidence score; higher is better.
gate3_translational_score / experimental_priority_score	Final experiment-priority scores; higher means better translational value.
tox_safety_score	Combined toxicity safety summary; higher means safer overall profile.
ddi_safety_score	Drug–drug interaction safety summary; higher means lower predicted DDI liability.
absorption_score	Combined absorption/permeability support; higher means better oral/transport potential.
hERG, DILI, ClinTox, AMES, Carcinogens_Lagunin	Risk probabilities/proxies; lower is safer because high values indicate more predicted toxicity concern.
max_cyp_inhibition	Maximum CYP inhibition risk across enzymes; lower is safer for DDI liability.
HIA_Hou, Bioavailability_Ma, PAMPA_NCATS	Absorption-related probabilities/scores; higher is better.
Caco2_Wang	Cell permeability proxy; less negative / higher means better permeability.

Simple reading rule for the final workbook

Prefer molecules with **high** final priority / safety / absorption scores, but **low** uncertainty, alert count, toxicity risk, CYP inhibition risk, and predicted IC_{50}.

Tools used in this repository: core modeling and generation

Tool	Used for in this repo
Python	Main language for the full pipeline, orchestration scripts, QSAR training, scoring, and reporting.
NumPy	Numeric arrays, descriptor matrices, statistics, and score calculations.
Pandas	Reading Excel/CSV inputs, manipulating scored tables, exporting candidate workbooks.
RDKit	SMILES parsing, canonicalization, descriptors, fingerprints, scaffolds, similarity, and structure rendering.
Mordred	Additional molecular descriptor generation for QSAR feature sets.
scikit-learn	Core QSAR regression models, preprocessing, comparison, and evaluation.
XGBoost / LightGBM / CatBoost	Gradient-boosting model families benchmarked as candidate QSAR predictors.
REINVENT4	Molecular generation engine for de novo, RL, CL, Mol2Mol, LibINVENT, LinkINVENT, and PepINVENT campaigns.

Tools used in this repository: scoring, visualization, and outputs

Tool	Used for in this repo
ADMET-AI	Predicted ADME/toxicity endpoints used in post-generation enrichment and shortlist annotations.
RDKit SA_Score	Synthetic accessibility estimation available for chemistry assessment features.
Matplotlib / Seaborn	Plots for QSAR distributions, cluster views, and reporting visuals.
Jupyter Notebooks	Exploratory analysis, debugging, and reproducible workflow inspection notebooks in the repo.
Excel / workbook export	Final ranked candidate review tables with structures for chemistry/biology decision-making.
LaTeX Beamer	This presentation deck and PDF reporting output.
External chemistry resources (for example `external_tools/SwissMADE`)	Supporting reference/tooling assets stored alongside the workflow.

Simple summary

The repo combines **cheminformatics** (RDKit, Mordred), **machine learning** (scikit-learn, boosting models), **molecular generation** (REINVENT4), and **developability scoring** (ADMET-AI + Gate 1 logic) into one pipeline.

What tools are used in the active Gate 1 workflow?

Gate 1 tools

- RDKit: molecule handling, descriptors, scaffold logic
- ADMET-AI: ADME/tox annotations used by scoring outputs
- QSAR ensemble utilities: potency prediction and uncertainty columns
- Pandas/NumPy: triage scoring, ranking, and shortlist selection
- `run_hybrid_stage_gates.py`: builds Gate 1 ranked outputs

Implementation file

Gate 1 is implemented in `reinvent_integration/run_hybrid_stage_gates.py`.

Important notes

Gate 2 and Gate 3 are not active in the current pipeline; older references remain only for provenance. Simulation Plus, GastroPlus, and ADMET Predictor are **not used** in the codebase and appear only as comparison examples in documentation.

Thank you

Questions?